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Abstract

Project Title: Agricare – Smart Agricultural E-Commerce & Advisory System

1. Abstract

Agriculture is the backbone of the economy, yet farmers often struggle with two critical challenges: limited access to quality agricultural inputs (seeds, fertilizers) and a lack of timely, expert technical advice for crop management. **Agricare** is a comprehensive web-based platform developed using the **Django** framework to bridge this gap.

The system serves a dual purpose: it functions as a niche **E-Commerce marketplace** where farmers can purchase high-quality seeds, pesticides, and fertilizers, and acts as an **AI-driven advisory consultant**. The core technical innovation lies in the integration of Machine Learning (ML) models directly into the web application. A **Plant Disease Recognition System** to diagnose crop ailments from uploaded leaf images, providing instant treatment suggestions. Simultaneously, the **Crop & Fertilizer Recommendation System** analyzes soil parameters (N, P, K, pH) and environmental data to scientifically recommend the most suitable crops and the precise fertilizer dosage required. By centralizing supply chain access and expert agronomy advice, Agricare empowers farmers to maximize crop yield, reduce input wastage, and make data-driven decisions.

2. Key Features

A. User/Farmer Module

- **Smart E-Commerce Store:**
 - Browse products by category (Seeds, Fertilizers, Pesticides, Machinery).

- Detailed product descriptions with usage guides.
- **Shopping Cart & Secure Checkout** functionality.
- Order History and Tracking.
- **AI Plant Disease Detection:**
 - **Image Upload:** Farmers can upload photos of infected leaves.
 - **Instant Diagnosis:** The system identifies the disease (e.g., "Tomato Late Blight") with a confidence score.
 - **Remedy Suggestion:** Automatically suggests specific pesticides or organic cures available in the store.
- **Intelligent Recommendations:**
 - **Crop Recommendation:** Input soil data (Nitrogen, Phosphorous, Potassium, pH) and location to get the best suitable crop suggestions.
 - **Fertilizer Recommendation:** Input the specific crop and soil nutrient levels to receive a calculated recommendation for fertilizer type and quantity.

B. Administrator Module

- **Product Management:** Add, update, or delete products, stocks, and prices.
 - **Order Management:** View, process, and update the status of user orders (Pending, Shipped, Delivered).
 - **AI Model Management:** Interface to update or retrain the disease detection models as new data becomes available.
 - **User Management:** Manage registered farmers and handle support queries.
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3. Technology Stack

Component	Technology Used	Description
Backend Framework	Django (Python)	Handles URL routing, database ORM, authentication, and core business logic.
Frontend	HTML5, CSS3, JavaScript	Structure and styling. Libraries like Bootstrap or Tailwind CSS are recommended for responsiveness.
Database	SQLite	SQLite is sufficient for development. Stores product data, user info, and orders.
Machine Learning	TensorFlow / Keras	For image-based disease detection.
Data Science	Scikit-Learn & Pandas	Used for the Crop/Fertilizer recommendation algorithms (e.g., Random Forest or Decision Tree classifiers).
Image Processing	OpenCV / Pillow	Handles image resizing and preprocessing before feeding images to the AI model.
Payment Gateway	Razorpay	For handling real-time payments during checkout.